

Assignment – 1

- i. List and briefly describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.**
- ii. What are features and needs of Machine Learning? Provide examples.**
- iii. Discuss 5 applications of Machine Learning with example.**
- iv. Describe Challenges in machine Learning.**
- v. What are overfitting and underfitting in Machine Learning models? How can they be prevented?**
- vi. What is Hypothesis Space and Inductive Bias? How does inductive bias relate to the hypothesis space? Explain with an example.**
- vii. What is Cross-Validation? Describe different types of cross-validation techniques, such as k-fold cross-validation and leave-one-out cross-validation.**

(NOTE: All students must Complete this question)